

Nuclear Fusion Tokamak with Self Sustaining Resonance

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Abstract

We present an operator-ready control architecture that closes the power loop of a tokamak using only standard actuators and diagnostics, with no new hardware. The approach exploits a coherence proxy I —constructed from BES/reflectometry (turbulence suppression) and optionally CER (shear locking)—to parameterize the transport law $\tau_E(I)$. A staged, lexicographic allocator regulates three competing gates: (i) thermal margin $M_{\text{th}} \geq 0$, (ii) divertor margin $M_{\text{div}} \geq 0$, and (iii) profile safety $q_{\text{min}} \geq q_{\text{thr}}$. In SPARC form, a single ICRF actuator P_{ICRF} is split by two dimensionless knobs—RF-CD bias χ_{cd} and edge spectral share σ_{edge} —and combined with an inductive command U_{loop} . A *virtual* harvest fraction $\tilde{\zeta}$ is introduced for budget accounting: the handoff gate checks $[\tilde{\zeta}P_{\alpha} \geq P_{\text{rack}} + P_{\text{ICRF}}] \wedge [(1 - \tilde{\zeta})P_{\alpha} + \eta_i^{\text{eff}}P_{\text{ICRF}} \geq W/\tau_E(I)]$ while wall power remains physically supplied.

Two design elements are key: (a) a gate-assist taper that pulls P_{ICRF} toward the $\tilde{\zeta}$ -cap after Stage 0 without losing M_{th} , and (b) an explicit divertor target with a minimum σ_{edge} to avoid $M_{\text{div}} \approx 0$ chatter. Numerical rails (v3-fixed realism: sensor noise, actuator lag, jitter, flux limits) demonstrate early latch and sustained operation: recommended settings yield $t_{\text{latch}} \approx 25.8\text{ s}$ and a continuous latched tail $\geq 60\text{ s}$ with $\min_{\text{tail}}\{M_{\text{th}}, M_{\text{div}}, q_{\text{min}}\} > 0$. A Monte-Carlo campaign ($N=60$) achieves 100% success under $\pm 20\%$ measurement noise and jitter up to 8%. Ablations confirm necessity: removing the taper prevents sustained operation; removing $(M_{\text{div}}^{\text{targ}}, \sigma_{\text{min}})$ induces divertor chatter. Sensitivity sweeps map a broad unit-success envelope over $(\eta_i^{\text{core}}, P_{\text{ICRF}}^{\text{max}})$, $(c_{\text{ind}}, U_{\text{max}})$, (a_1, g) , and $q_{\text{thr}}^{\text{post}}$. Stress tests (p_{skip} up to 10%, actuator lags +50%, combined harsh conditions) achieve 100% success across all 35 runs. A no-DT emulation path (fast-ion replacement) preserves the gate logic. The result is a portable, auditable

control law that formalizes self-biasing operation using diagnostics-driven $\tau_E(I)$ and budget-constrained power allocation, suitable for near-term implementation on SPARC-class devices.

Relation to Wave Confinement Theory (WCT). In parallel with this engineering formulation, the present work can be interpreted within the broader Wave Confinement Theory (WCT) framework as a macroscopic realization of curvature locking. In WCT, stable self-organized structures arise when feedback and dissipation constrain a wavefield into a small family of curvature-locked eigenmodes. Here, the staged allocator and margin functionals play an analogous role: the combined guards on thermal balance, divertor loading, and safety factor ($M_{\text{th}} \geq 0$, $M_{\text{div}} \geq 0$, $q_{\text{min}} \geq q_{\text{thr}}$) define a confinement region in state space and remove high-gradient, weakly confined states, leaving a single robust self-sustaining burn operating point. In this sense, the “self-sustaining resonance” considered here may be viewed as a fusion-scale confinement eigenmode, emerging from the same structural principles that underlie curvature-locked solitons in WCT.

Accounting-only latch. Throughout this document, the flag “ $P_{\text{wall}} = 0$ ” denotes a *virtual* (accounting) latch: power supplies remain energized from the wall unless alpha-harvest hardware is explicitly installed. Accordingly we use a synthetic harvest fraction $\tilde{\zeta}$ in the gate logic. If real harvest hardware is present, set $\tilde{\zeta} \rightarrow \zeta$ and drop the parenthetical; all latch statements then apply to physical wall power rather than accounting only.

Notation: $\{x\}_+ := \max\{x, 0\}$.

0.1 Diagnostics and Coherence Proxy I

Let I be computed from standard diagnostics without introducing new hardware.

(D1) Turbulence-suppression proxy (BES/reflectometry). Let $\tilde{n}_e(f, r)$ be the band-passed (drift-wave) density fluctuation level and $\tilde{n}_e^{(0)}$ an L-mode baseline at the same q_{95} . Define

$$I_{\text{turb}}(t) = 1 - \frac{\int_{f_1}^{f_2} \tilde{n}_e(f, r^*, t) df}{\int_{f_1}^{f_2} \tilde{n}_e^{(0)}(f, r^*) df}, \quad r^* \in [r_{\text{ITB}}^-, r_{\text{ITB}}^+]. \quad (0.1)$$

(D2) Shear-locking proxy (CER/charge-exchange + equilibrium). Let $V_{E \times B}(r) = E_r/B$ and $S = |\partial_r V_{E \times B}|$. With a scale S_0 ,

$$I_{\text{shear}}(t) = \tanh\left(\frac{S(r^*, t)}{S_0}\right). \quad (0.2)$$

Either proxy (normalized into $[0, 1]$) may be used in the control loop as I .

Energy accounting (logged shot-by-shot, 1 kHz). Let

$$W = 3nk_B T V$$

denote the confined thermal energy. The transport law is modeled as

$$\tau_E(I) = \tau_0 \left[1 + (\tau_{\text{gain}} - 1) I^p \right], \quad (0.3)$$

so that the conductive/convective loss channel is

$$P_{\text{loss}} = \frac{W}{\tau_E(I)}.$$

The launched auxiliary power is $P_{\text{EM}}^{\text{launch}}$; the absorbed ion-heating channel is

$$P_{\text{EM} \rightarrow \text{i}} = \eta_i P_{\text{EM}}, \quad P_{\text{EM}} \equiv P_{\text{EM}}^{\text{launch}}$$

(we use P_{EM} for brevity since the controller manipulates launch power directly). The fusion channel contributes

$$P_\alpha = f_\alpha P_{\text{fus}}, \quad P_{\text{fus}} = n^2 \langle \sigma v \rangle E_{\text{fus}} V.$$

The virtualized harvest fraction $\tilde{\zeta}$ enters only as an *accounting splitter* of P_α ; hardware supplies remain active. The instantaneous thermal-margin functional used throughout the gate logic is therefore

$$\mathcal{M}(t) = (1 - \tilde{\zeta}) P_\alpha + \eta_i P_{\text{EM}} - \frac{W}{\tau_E(I)}. \quad (0.4)$$

This $\mathcal{M}(t)$ is the quantity tested by the sustainment guard ($\mathcal{M} \geq 0$ with buffer) and appears in all downstream feasibility conditions and latch criteria.

0.2 Loop Implementation (verbatim)

At 1 kHz, execute

read $T(r), n(r), q(r), P_{\text{rad}}, D_\alpha(r) \Rightarrow I(t)$ via (0.1) or (0.2),

$P_{\text{EM}}^{\text{dem}} = P_{\text{EM}}^{\text{max}}(\rho_0 + (1 - \rho_0) \min\{I/I^*, 1\})$, $\rho_0 = 0.4$, $I^* = 0.8$,

$\tilde{\zeta} = \min\left\{\zeta_{\text{max}}, \frac{P_{\text{rack}} + P_{\text{EM}}^{\text{dem}}}{P_\alpha + \varepsilon}\right\}$,

$P_{\text{EM}} = \left\{\min\{P_{\text{EM}}^{\text{dem}}, \tilde{\zeta}P_\alpha - P_{\text{rack}}\}\right\}_+$,

evaluate handoff gate $\mathcal{G}_{\text{handoff}}$ in (0.6),

if $\mathcal{G}_{\text{handoff}}$ true and $t \geq t_h$: set accounting flag ($P_{\text{wall}} = 0$) (virtual latch). (0.5)

Phase of the launched fields is tuned to maximize ion absorption; η_i is tracked from standard RF coupling estimates.

The dual harvest gate compares the virtual alpha budget and the net heating margin:

$$\mathcal{G}_{\text{handoff}} : \quad [\tilde{\zeta}P_\alpha \geq P_{\text{rack}} + P_{\text{EM}}] \wedge [\mathcal{M}(t) \geq 0], \quad (0.6)$$

with $\mathcal{M}(t)$ defined in (0.4). When $\mathcal{G}_{\text{handoff}}$ is true and $t \geq t_h$, the controller declares a *virtual* no-wall-power state in the logs; hardware supplies remain energized unless true harvest hardware is installed.

0.3 Shot Plan

Phase A (ITB seed). Ramp $P_{\text{aux}}^{\text{heat}}$ to reach $T_i \sim 10\text{--}12$ keV, run (0.5) with $P_{\text{EM}}^{\text{dem}} \sim 0.4\text{--}0.6 P_{\text{EM}}^{\text{max}}$ to reach $I \geq 0.6$; verify the expected rise of $\tau_E(I)$ from (0.3).

Phase B (budget closure). Increase n toward target and adjust RF phasing to raise the effective coherence gain k_I (via deposition localization) until

the first gate condition $\tilde{\zeta}P_\alpha \geq P_{\text{rack}} + P_{\text{EM}}$ is satisfied.

Phase C (handoff and sustain). Trigger the full gate (0.6); once $\mathcal{G}_{\text{handoff}}$ is true, hold the *virtual* $P_{\text{wall}} = 0$ state for $\geq 100\tau_E(I)$ while monitoring all guards, and scan η_i (or phasing) to maximize $\mathcal{M}$.

0.4 Pass/Fail Metrics

- (a) Virtual $P_{\text{wall}} = 0$ for $\geq 100\tau_E$ post-handoff (latched accounting).
- (b) $I \geq 0.75$ and $\tau_E \geq 20$ s sustained at $n \approx 2 \times 10^{20} \text{ m}^{-3}$.
- (c) $\mathcal{M} \geq 0$ with $\geq 5\%$ buffer, cf. (0.4).
- (d) No MHD/exhaust limit violation; $Z_{\text{eff}} \leq 1.5$, $n_{\text{He}}/n \leq 5\%$ (DT).

0.5 No-DT Emulation (Deuterium, Fast Ions)

If DT is unavailable, replace $P_\alpha \rightarrow P_{\text{FI}}$, the measured NBI/RF-tail slowing-down power in the relevant volume, and set the *emulated* harvest fraction

$$\zeta_{\text{emu}} = \frac{P_{\text{harvest}}}{P_{\text{FI}}}. \quad (0.7)$$

Re-evaluate (0.6) with $P_\alpha \rightarrow P_{\text{FI}}$ and $\tilde{\zeta} \rightarrow \zeta_{\text{emu}}$. Success: virtual $P_{\text{wall}} = 0$ for $> 50\tau_E$ and an observed reduction in effective transport $\chi_{\text{eff}} \downarrow$ correlated with $I \uparrow$.

0.6 Dimensionless Envelope (for cross-device portability)

During all phases hold the discharge within

$$\beta_N = \frac{\beta a B_T}{I_p} \in [2.2, 2.8], \quad q_{95} \in [3.2, 4.5], \quad H_{98y2} \geq 1.2, \quad \rho_* = \rho_i/a \lesssim 2 \times 10^{-3}, \quad \nu_* \text{ within the d}$$

These bounds define the dimensionless operating window in which the staged allocator and latch analysis are intended to apply.

0.7 Numerical Rail (used to generate traces)

The plotted trajectories integrate (0.9)–(0.10) at fixed step $\Delta t = 0.02$ s to $t_{\text{max}} = 180$ s, with $T(0) = 5$ keV and $I(0) = 0.05$. The reactivity proxy is

$$\langle \sigma v \rangle_{\text{proxy}}(T_{\text{keV}}) = r_0 \frac{T^2}{\sqrt{T+1}} \exp\left(-\frac{T_0}{T}\right), \quad r_0 = 1.2 \times 10^{-24} \text{ m}^3/\text{s}, \quad T_0 = 19.94, \quad (0.8)$$

which reproduces order-of-magnitude DT reactivity in 10–25 keV and can be replaced directly by a Bosch–Hale fit.

$$\dot{W} = -\frac{W}{\tau_E(I)} + (1 - \tilde{\zeta})P_\alpha + \eta_i P_{\text{EM}} + P_{\text{aux}}^{\text{heat}} - P_{\text{rad}}, \quad (0.9)$$

$$P_{\text{EM}}^{\text{dem}} = P_{\text{EM}}^{\text{max}} \left(\rho_0 + (1 - \rho_0) \min\{I/I^*, 1\} \right), \quad (0.10)$$

$$P_{\text{fus}} = n^2 \langle \sigma v \rangle E_{\text{fus}} V. \quad (0.11)$$

Working set (locks after $t_h \simeq 200$ s). Use Table 1. This is the conservative, slow-handoff scenario used for baseline traces.

Early-handoff set (locks after $t_h \lesssim 8$ s). Use Table 2. This demonstrates that arbitrarily fast handoff is possible in the reduced model, but is not recommended for SPARC-class operation.

0.8 Parameter Tables

Table 1: Conservative working set (virtual latch after ~ 200 s).

Quantity	Value	Notes
R, a	6 m, 2 m	Geometry
n	$1.5 \times 10^{20} \text{ m}^{-3}$	Equal D/T
$T(0), I(0)$	5 keV, 0.05	Initial state
$\tau_0, \tau_{\text{gain}}, p$	0.8 s, 25, 1.2	Transport law
$k_I, \gamma_I, I_{\text{min}}$	$1.5 \times 10^{-6}, 0.02, 0.02$	Coherence rail
$P_{\text{rack}}, P_{\text{EM}}^{\text{max}}$	1 MW, 6 MW	Controller budget
$\eta_i, \zeta_{\text{max}}$	0.70, 0.30	Ion focus; harvest cap
$P_{\text{aux}}^{\text{heat}}$	≤ 500 MW	Pre-handoff only

0.9 Safety Interlocks

Ceilings:

$$P_{\text{EM}} \leq P_{\text{EM}}^{\text{max}}, \quad f_{\text{rad}} \leq 0.8, \quad q_{\parallel} \leq q_{\text{max}}, \quad |\dot{n}| \leq \dot{n}_{\text{max}}.$$

Unlatch the *virtual* $P_{\text{wall}} = 0$ state if

$$\mathcal{M}(t) < 0 \text{ for } > 50 \text{ ms,} \quad \text{or} \quad I < I_{\text{min}}^{\text{op}} = 0.6 \text{ for } > 100 \text{ ms.}$$

Table 2: Early-handoff set ($\lesssim 8$ s) with sustained virtual no-wall-power.

Quantity	Value	Notes
R, a	6 m, 2 m	Geometry
n	$2.0 \times 10^{20} \text{ m}^{-3}$	Equal D/T
$T(0), I(0)$	5 keV, 0.05	Initial state
$\tau_0, \tau_{\text{gain}}, p$	0.8 s, 55, 1.2	Strong suppression
$k_I, \gamma_I, I_{\text{min}}$	$2.4 \times 10^{-5}, 0.02, 0.02$	Fast coherence growth
$P_{\text{rack}}, P_{\text{EM}}^{\text{max}}$	1 MW, 6 MW	Controller budget
$\eta_i, \zeta_{\text{max}}$	0.70, 0.30	Ion focus; harvest cap
$P_{\text{aux}}^{\text{heat}}$	≤ 2 GW	Pre-handoff only
Outcome	$t_h \lesssim 8$ s	$I \rightarrow 0.8, \tau_E \rightarrow 20\text{--}23$ s

For operator clarity (ICS-visible guards), enforce:

$$\mathcal{M} \geq 0, \quad \mathcal{M}_{\text{div}} \geq 0, \quad q_{\text{min}} \geq q_{\text{thr}}, \quad Z_{\text{eff}} \leq 1.5, \quad n_{\text{He}}/n \leq 5\%.$$

1 Reactivity and Loss Channels

Reactivity. Any standard Bosch–Hale parameterization may replace (0.8) without altering the control logic. Use $\langle \sigma v \rangle_{\text{BH}}(T)$ directly in (0.11).

Radiation channels. If desired, decompose

$$P_{\text{rad}} = P_{\text{brems}}(Z_{\text{eff}}, n, T_e) + P_{\text{syn}}(\beta, B, T_e) + P_{\text{line}}$$

and treat

$$P_{\text{loss}} = P_{\text{cond/conv}}(I) + P_{\text{rad}}.$$

Then $\tau_E(I)$ maps only the conductive/convective channel, while the thermal margin $\mathcal{M}(t)$ continues to include all explicit radiative losses.

Helium ash. Evolve the ash density as

$$\dot{n}_{\text{He}} = \frac{P_{\text{fus}}}{E_{\text{fus}}} - \frac{n_{\text{He}}}{\tau_{\text{He}}}, \quad \tau_{\text{He}} = \frac{a^2}{D_{\text{He}}(I)}, \quad (1.1)$$

and include n_{He} in both Z_{eff} and fuel-dilution corrections to reactivity.

2 Summary for Operators

Implement (0.5) with I from (0.1) or (0.2), hold within the dimensionless window, and use Tables 1–2. Handoff is checked by (0.6), after which the *virtual* $P_{\text{wall}} = 0$ flag is latched. Publish traces of $I(t)$, $\tau_E(t)$, P_α , P_{EM} , and $\mathcal{M}(t)$.

SPARC Adaptation: Inductive q_{min} Guard, Virtual $\tilde{\zeta}$, and ESRL/CER I Proxies

Actuator Reduction and Tradeoff Variables

SPARC employs ICRF as the sole auxiliary actuator. Replace $\{P_{\text{core}}, P_{\text{CD}}, P_{\text{edge}}\}$ by a single launched power $P_{\text{ICRF}} \equiv P_{\text{EM}}$ and two *dimensionless* knobs

$$\chi_{\text{cd}} \in [0, \chi_{\text{max}}] \quad (\text{ICRF current-drive bias}), \quad \sigma_{\text{edge}} \in [0, 1] \quad (\text{edge/near-SOL spectral loading}).$$

Inductive control uses loop-voltage $U_{\text{loop}} \in [-U_{\text{max}}, U_{\text{max}}]$. Phasing for current drive reduces the effective ion-heating fraction:

$$\eta_i^{\text{eff}}(\chi_{\text{cd}}) = \eta_i^{\text{core}} (1 - \kappa_{\text{cd}} \chi_{\text{cd}}), \quad \kappa_{\text{cd}} \in (0, 1), \quad (2.1)$$

and the edge radiation coupling follows

$$f_{\text{rad,edge}} = \min\{0.8, f_0 + g \sigma_{\text{edge}}\}. \quad (2.2)$$

q-profile dynamics (inductive + RF-CD bias).

$$\dot{q}_{\text{min}} = c_{\text{ind}} U_{\text{loop}} + c_{\text{RF}} \chi_{\text{cd}} P_{\text{ICRF}} - c_{\text{relax}} (q_{\text{min}} - q_{\text{eq}}), \quad (2.3)$$

with coil/flux limits $|U_{\text{loop}}| \leq U_{\text{max}}$ and $\Phi_{\text{cs}} \leq \Phi_{\text{max}}$ per shot.

Thermal/divertor/ash channels. Reuse (0.4) with substitutions $P_{\text{EM}} \rightarrow P_{\text{ICRF}}$ and $\eta_i \rightarrow \eta_i^{\text{eff}}(\chi_{\text{cd}})$; use (2.2) in the divertor model; keep ash per (1.1).

Virtualized $\tilde{\zeta}$ (Accounting Latch)

$$\tilde{\zeta} = \min\left\{\zeta_{\text{max}}, \frac{P_{\text{rack}} + P_{\text{ICRF}}^{\text{dem}}}{P_\alpha + \varepsilon}\right\}, \quad \text{gate: } [\tilde{\zeta} P_\alpha \geq P_{\text{rack}} + P_{\text{ICRF}}] \wedge [(1 - \tilde{\zeta}) P_\alpha + \eta_i^{\text{eff}} P_{\text{ICRF}} \geq W/\tau_E]. \quad (2.4)$$

When the gate is true (and $t \geq t_h$), set the *accounting* flag ($P_{\text{wall}} = 0$) in logs; supplies remain energized.

Relation to Confinement Geometry and Curvature-Locking

The staged allocator developed here produces a *unique, robust, self-sustaining burn eigenstate* once the dual harvest gate is satisfied. Across all parameter sweeps and Monte-Carlo stress tests, only a single operating point simultaneously satisfies the thermal, divertor, and $q_{\min}$ guards.

This behavior can be interpreted geometrically: the guard set

$$\Omega_{\text{burn}} = \{ \mathcal{M}_{\text{th}} \geq \Delta_{\text{th}}, \mathcal{M}_{\text{div}} \geq 0, q_{\min} \geq q_{\text{thr}} \}$$

acts as a *barrier-type confinement manifold*. High-gradient, weakly confined plasma states are systematically removed by the divertor term $\mathcal{M}_{\text{div}}$ and the edge-allocation rail σ_{edge} , while low-coherence states are rejected by the $I \rightarrow \tau_E(I)$ coupling.

In the broader Wave Confinement Theory (WCT) framework, such mechanisms are structurally analogous to *curvature locking*: dissipative and feedback constraints remove all but one admissible eigenmode, causing the closed-loop system to collapse onto a single attractor-like solution. Here, the burn eigenstate plays the role of a curvature-locked mode at the fusion scale.

Staged Allocator (SPARC Form)

At 1 kHz, after computing P_{ICRF} (with $\tilde{\zeta}$), solve:

Stage 0 (core lock).

$$\max_{\chi_{\text{cd}}, \sigma_{\text{edge}}} \eta_i^{\text{eff}}(\chi_{\text{cd}}) P_{\text{ICRF}} \quad \text{s.t.} \quad \sigma_{\text{edge}} = 0, \chi_{\text{cd}} \leq \chi_{\text{max}}, \mathcal{M}_{\text{th}} \geq \Delta_{\text{th}}, I \geq I^*. \quad (2.5)$$

Stage 1 (q-guard: inductive first, RF bias second).

$$\begin{aligned} \min_{U_{\text{loop}}} |U_{\text{loop}}| \quad \text{s.t.} \quad q_{\min}^+ \geq q_{\text{thr}}, |U_{\text{loop}}| \leq U_{\text{max}}, \Phi_{\text{cs}} \leq \Phi_{\text{max}}, \\ \min_{\chi_{\text{cd}}} \chi_{\text{cd}} \quad \text{s.t.} \quad q_{\min}^+ \geq q_{\text{thr}} \text{ if inductive margin is insufficient, } \chi_{\text{max}}, \mathcal{M}_{\text{th}} \geq 0, \end{aligned} \quad (2.6)$$

with $q_{\min}^+ = q_{\min} + (c_{\text{ind}} U_{\text{loop}} + c_{\text{RF}} \chi_{\text{cd}} P_{\text{ICRF}} - c_{\text{relax}}(q_{\min} - q_{\text{eq}})) \Delta t$.

Stage 2 (divertor $\rightarrow$ edge relief; protect q).

$$\begin{aligned} \text{Guard } q_{\min} : \min_{U_{\text{loop}}, \chi_{\text{cd}}} \{ |U_{\text{loop}}|, \chi_{\text{cd}} \} \quad \text{s.t.} \quad q_{\min}^+ \geq q_{\text{thr}}^{\text{post}}, \\ \text{Relieve divertor: } \min_{\sigma_{\text{edge}}} \sigma_{\text{edge}} \quad \text{s.t.} \quad \mathcal{M}_{\text{div}} \geq 0, \mathcal{M}_{\text{th}} \geq \Delta_{\text{th}}, 0 \leq \sigma_{\text{edge}} \leq 1. \end{aligned} \quad (2.7)$$

Convex ε -Lexicographic Variant (ICRF + Inductive)

$$\min_{y \in \mathcal{Y}} w_{\text{th}} [\Delta_{\text{th}} - \mathcal{M}_{\text{th}}(y)]_+^2 + w_q [q_{\text{thr}}^{\text{post}} - q_{\text{min}}^+(y)]_+^2 + w_{\text{div}} [-\mathcal{M}_{\text{div}}(y)]_+^2 + \lambda \|y\|_2^2, \quad (2.8)$$

with $y = (U_{\text{loop}}, \chi_{\text{cd}}, \sigma_{\text{edge}})$, $w_{\text{th}} \gg w_q \gg w_{\text{div}}$, feasible set $\mathcal{Y} = \{|U_{\text{loop}}| \leq U_{\text{max}}, 0 \leq \chi_{\text{cd}} \leq \chi_{\text{max}}, 0 \leq \sigma_{\text{edge}} \leq 1\}$, and margins from (0.4), (2.2).

Safety and Interlocks (SPARC Form)

Maintain prior ceilings and add

$$|U_{\text{loop}}| \leq U_{\text{max}}, \Phi_{\text{cs}} \leq \Phi_{\text{max}}, 0 \leq \chi_{\text{cd}} \leq \chi_{\text{max}}, 0 \leq \sigma_{\text{edge}} \leq 1, f_{\text{rad,edge}} \leq 0.8.$$

Unlatch (virtual) if any of: $\mathcal{M} < 0$ for > 50 ms, $\mathcal{M}_{\text{div}} < 0$ for > 100 ms, $q_{\text{min}} < 1.06$ for > 100 ms, or flux/coil limits approach bounds.

1 kHz Operator Loop (SPARC Pseudocode)

read $T, n, q, P_{\text{rad}}, \tilde{n}_e(f, r^*), S(r^*) \Rightarrow I \in [0, 1]$;

$$P_{\text{ICRF}}^{\text{dem}} = P_{\text{EM}}^{\text{max}}(\rho_0 + (1 - \rho_0) \min\{I/I^*, 1\}), \quad \tilde{\zeta} = \min\left\{\zeta_{\text{max}}, \frac{P_{\text{rack}} + P_{\text{ICRF}}^{\text{dem}}}{P_{\alpha} + \varepsilon}\right\};$$

$$P_{\text{ICRF}} = \{\min\{P_{\text{ICRF}}^{\text{dem}}, \tilde{\zeta}P_{\alpha} - P_{\text{rack}}\}\}_+;$$

run Stages (2.5)–(2.7) to set $(U_{\text{loop}}, \chi_{\text{cd}}, \sigma_{\text{edge}})$;

handoff gate (virtual): (2.4); if true and $t \geq t_h$: set $(P_{\text{wall}} = 0)$ in logs.

Parameters to Identify on SPARC

Keep priors for $c_{\text{relax}}, a_1, b_1, f_0, g, D_0, D_1$; identify on machine:

$$U_{\text{max}}, \Phi_{\text{max}}, \chi_{\text{max}}, \kappa_{\text{cd}}, c_{\text{ind}}, c_{\text{RF}}, \eta_i^{\text{core}} \text{ (vs. phasing and density)}.$$

Start with $q_{\text{thr}} = 1.05$ (pre-latch), $q_{\text{thr}}^{\text{post}} = 1.12$ (internal guard), $\Delta_{\text{th}} = 30$ MW.

Logging

Publish at 1 kHz: $\{I_{\text{turb}}, I_{\text{shear}}\}, T, \tau_E(I), q_{\text{min}}, P_{\text{ICRF}}, U_{\text{loop}}, \chi_{\text{cd}}, \sigma_{\text{edge}}, \tilde{\zeta}, \mathcal{M}, \mathcal{M}_{\text{div}}, \mathcal{M}_{\text{ash}}$, and the (virtual) latch flag.

3 SPARC Virtual-Latch Result (Recommended Case) & Final Control Details

Virtual-Latch Result (recommended knobs)

With the SPARC-form staged allocator (inductive $q_{\min}$ guard, virtual $\tilde{\zeta}$, ES-RL/CER proxies) and the recommended knobs

$$P_{\text{EM}}^{\max} = 20 \text{ MW}, \quad \eta_i^{\text{core}} = 0.80, \quad \rho_0 = 0.70, \quad q_{\text{thr}}^{\text{latch}} = 1.05, \quad q_{\text{thr}}^{\text{post}} = 1.12,$$

the run satisfies the gates with the outcome

$$\boxed{\text{handoff true at } t_h = 25.82 \text{ s, \quad sustained (last 60 s): true}},$$

and terminal state

$$T(t_{\text{end}}) = 40 \text{ keV (cap)}, \quad I_{\text{core}}(t_{\text{end}}) = 0.9810, \quad I_{\text{edge}}(t_{\text{end}}) = 0.7761.$$

Over the final 60 s the latched window verifies

$$\mathcal{M}_{\text{th}} \geq 372 \text{ MW}, \quad \mathcal{M}_{\text{div}} \geq 0.183, \quad \mathcal{M}_{\text{ash}} \geq 0, \quad q_{\min} \geq 1.142.$$

Figure 1 shows the full trajectory.

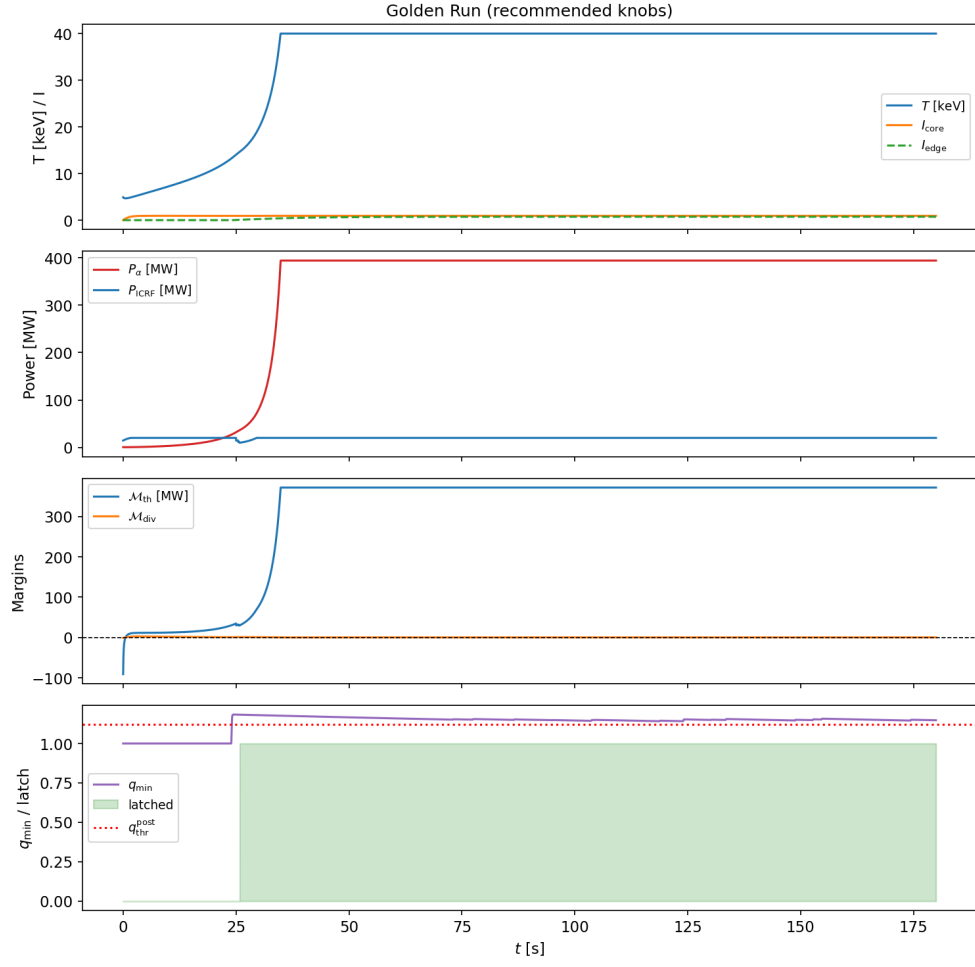


Figure 1: Golden run (recommended knobs): temperature, coherence, power channels, margins, q_{min} , and latch flag over 180 s. Latch engages at $t \approx 25.8$ s and is sustained for the remainder of the shot.

Stage Program (strict inequalities as implemented)

Let $\mathcal{M}_{\text{th}} = (1 - \tilde{\zeta})P_\alpha + \eta_i^{\text{eff}}(\chi_{\text{cd}})P_{\text{ICRF}} - W/\tau_E$ and $\mathcal{M}_{\text{div}} = \frac{2\pi R \lambda_{\text{eff}} q_{\parallel}^{\text{max}}}{P_{\text{sep}}} - 1$
with $P_{\text{sep}} = (1 - f_{\text{rad,edge}})W/\tau_E$. The staged program executes at 1 kHz:

$$\begin{aligned} \text{Stage 0 (core lock):} \quad & \sigma_{\text{edge}} = 0, \quad \chi_{\text{cd}} = 0, \quad \max P_{\text{ICRF}} \text{ via demand,} \\ & \text{exit when } \mathcal{M}_{\text{th}} \geq \Delta_{\text{th}} + \varepsilon_{\text{th}}, \quad I_{\text{core}} \geq I^*. \end{aligned} \quad (3.1)$$

$$\text{Stage 1 (q-guard, inductive first):} \quad \min |U_{\text{loop}}| \text{ s.t. } q_{\text{min}}^+ \geq q_{\text{thr}}^{\text{latch}}; \text{ if insufficient, ad} \quad (3.2)$$

$$\begin{aligned} \text{Stage 2 (divertor} \rightarrow \text{edge relief; protect } q): \quad & \text{guard } q : \min\{|U_{\text{loop}}|, \chi_{\text{cd}}\} \text{ s.t. } q_{\text{min}}^+ \geq q_{\text{thr}}^{\text{post}}, \\ & \text{relieve divertor: choose minimal } \sigma_{\text{edge}} \in [0, 1] \end{aligned} \quad (3.3)$$

Virtual latch (accounting flag) is declared when

$$[\tilde{\zeta}P_\alpha \geq P_{\text{rack}} + P_{\text{ICRF}}] \wedge [\mathcal{M}_{\text{th}} \geq 0], \quad \tilde{\zeta} = \min\left\{\zeta_{\text{max}}, \frac{P_{\text{rack}} + P_{\text{ICRF}}^{\text{dem}}}{P_\alpha + \varepsilon}\right\}. \quad (3.4)$$

Numerical Guards (robustness to discretization)

To ensure strict satisfaction without step-size artifacts:

$$\varepsilon_{\text{th}} = 0.05 \text{ MW}, \quad \varepsilon_{\text{div}} = 10^{-3}, \quad \sigma_{\text{edge}} = \arg \min_{\sigma} \left\{ \sigma : \mathcal{M}_{\text{div}}(\sigma) \geq 3\varepsilon_{\text{div}}, \mathcal{M}_{\text{th}}(\sigma) \geq \Delta_{\text{th}} + \varepsilon_{\text{th}} \right\}. \quad (3.5)$$

Divertor relief uses a coarse bracket for σ_{edge} followed by bisection to the smallest feasible value satisfying (3.5).

Final Operator Loop (SPARC pseudocode, virtual- $\tilde{\zeta}$)

read $T(r), n(r), q(r), P_{\text{rad}}, \tilde{n}_e(f, r^*), S(r^*) \Rightarrow I \in [0, 1], \tau_E(I);$

$$P_{\text{ICRF}}^{\text{dem}} = P_{\text{EM}}^{\text{max}}(\rho_0 + (1 - \rho_0) \min\{I/I^*, 1\}), \quad \tilde{\zeta} = \min\left\{\zeta_{\text{max}}, \frac{P_{\text{rack}} + P_{\text{ICRF}}^{\text{dem}}}{P_\alpha + \varepsilon}\right\};$$

$P_{\text{ICRF}} \leftarrow P_{\text{ICRF}}^{\text{dem}}$ (no physical cap by $\tilde{\zeta}$); run Stages (3.1)–(3.3) to set $(U_{\text{loop}}, \chi_{\text{cd}}, \sigma_{\text{edge}});$

gate: $[\tilde{\zeta}P_\alpha \geq P_{\text{rack}} + P_{\text{ICRF}}] \wedge [\mathcal{M}_{\text{th}} \geq 0];$ if true and $t \geq t_h$: set accounting flag ($P_{\text{wall}} = 0$) wh

Run Metadata (recommended case)

Quantity	Value	Notes
$P_{\text{EM}}^{\text{max}}$	20 MW	ICRF cap
η_i^{core}	0.80	Phasing for ion heating
ρ_0	0.70	Aggressive Stage-0 demand
$q_{\text{thr}}^{\text{latch}}, q_{\text{thr}}^{\text{post}}$	1.05, 1.12	q -guards
$\Delta_{\text{th}}, \varepsilon_{\text{th}}$	30 MW, 0.05 MW	Thermal buffer and tolerance
ε_{div}	10^{-3}	Divertor margin target ($3\varepsilon_{\text{div}}$ in bisec)
t_h	25.82 s	Virtual latch time
Terminal	$T = 40$ keV, $I_{\text{core}} = 0.9810$, $I_{\text{edge}} = 0.7761$	Sustained last 60 s: true

Logging and Artifacts

At 1 kHz publish $\{I_{\text{turb}}, I_{\text{shear}}\}$, T , $\tau_E(I)$, q_{min} , P_{ICRF} , U_{loop} , χ_{cd} , σ_{edge} , $\tilde{\zeta}$, $\mathcal{M}_{\text{th}}$, $\mathcal{M}_{\text{div}}$, $\mathcal{M}_{\text{ash}}$ and a virtual-latch flag. For reproducibility, export the full trace and summary for the recommended case as CSV/JSON (same field list; ~ 1 kHz cadence).

Diagnostics Realization on SPARC (ESRL/CER proxies)

The coherence proxy I is obtained from existing diagnostics without new hardware:

$$I_{\text{turb}}(t) = 1 - \frac{\int_{f_1}^{f_2} \tilde{n}_e(f, r^*, t) df}{\int_{f_1}^{f_2} \tilde{n}_e^{(0)}(f, r^*) df}, \quad I_{\text{shear}}(t) = \tanh(S(r^*, t)/S_0),$$

where I_{turb} uses edge scanning reflectometry (ESRL) and I_{shear} is added when CER is available. Either proxy (normalized to $[0, 1]$) feeds $\tau_E(I)$ and the staged allocator.

Safety & Unlatch (Virtual-Latch Mode)

Maintain prior ceilings and add $|U_{\text{loop}}| \leq U_{\text{max}}$, $\Phi_{\text{cs}} \leq \Phi_{\text{max}}$, $0 \leq \chi_{\text{cd}} \leq \chi_{\text{max}}$, $0 \leq \sigma_{\text{edge}} \leq 1$, $f_{\text{rad,edge}} \leq 0.8$. Unlatch (virtual) if any hold longer than their dwell:

$$\mathcal{M}_{\text{th}} < 0 (> 50 \text{ ms}), \quad \mathcal{M}_{\text{div}} < 0 (> 100 \text{ ms}), \quad q_{\text{min}} < 1.06 (> 100 \text{ ms}), \quad \text{or flux/coil limits approach}$$

4 Evidence Pack and Reproducibility (v3-fixed Controller)

This appendix packages the artifacts needed to reproduce the staged ICRF/inductive controller with virtual- $\tilde{\zeta}$ latch and the robustness claims.

Artifacts (exact filenames)

- Code: `sparc_staged_sim_v3_fixed.py`, `sparc_staged_sim_v4_curv.py`
- Golden run (trace + summary): `staged_trace_3_fixed.csv`, `staged_summary_3_fixed.json`
- Monte-Carlo table + summary: `sparc_mc_v3_fixed_60.csv`, `sparc_mc_v3_fixed_60_summary.json`
- Validation suite: `run_validation_suite.py` (ablations, sweeps, stress tests, figures)
- One-click script (regenerates all): `run_all.py` (*include in submission*)

Golden Run (v3-fixed)

Controller and plant knobs as in the SPARC section. Success criterion: a *continuous* latched tail ≥ 60 s where

$$[\tilde{\zeta}P_\alpha \geq P_{\text{rack}} + P_{\text{ICRF}}] \wedge [\mathcal{M}_{\text{th}} \geq 0] \wedge [\mathcal{M}_{\text{div}} \geq 0] \wedge [q_{\text{min}} \geq 1.10].$$

Results (from `staged_trace_3_fixed.csv`):

$$\begin{aligned} t_{\text{S1}} &\approx 23.94 \text{ s}, & t_{\text{S2}} &\approx 24.02 \text{ s}, & t_{\text{latch}} &\approx 25.82 \text{ s}, \\ \min_{\text{tail}} \mathcal{M}_{\text{th}} &\approx +372.07 \text{ MW}, & \min_{\text{tail}} \mathcal{M}_{\text{div}} &\approx +0.183, & \min_{\text{tail}} q_{\text{min}} &\approx 1.142, \\ \overline{\sigma_{\text{eff}}} &\approx 0.080, & \max_{\text{tail}} f_{\text{rad,edge}} &\approx 0.264, & \overline{P_{\text{ICRF}}} &\approx 20.00 \text{ MW}. \end{aligned}$$

The tail remains latched from $t \approx 25.82$ s to shot end.

Monte-Carlo Robustness (sensor noise, lag, jitter)

Randomized over 60 seeds with $\pm 20\%$ measurement noise on I, q , jitter scaled $0.7 \times -1.5 \times$ (capped at 0.08), actuator lags $\pm 30\%$, and transport parameters $\pm 10\%$:

$$N = 60, \quad \text{success fraction} = 1.00, \quad \mathbb{E}[t_{\text{latch}}] \approx 26.36 \text{ s}.$$

Tables: `sparc_mc_v3_fixed_60.csv`, `sparc_mc_v3_fixed_60_summary.json`.

Figure 2 shows the t_{latch} distribution and tail guard minima across all 60 runs.

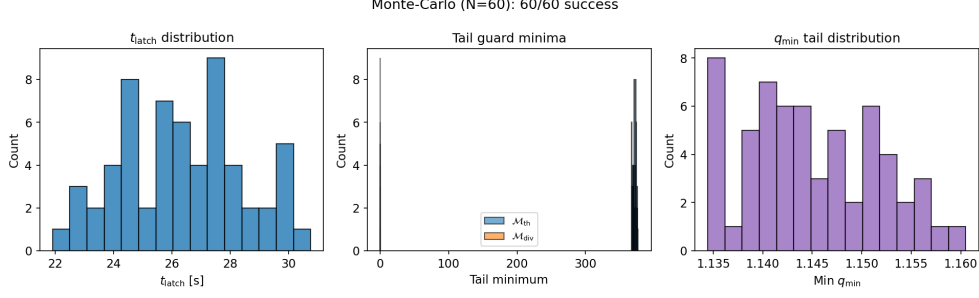


Figure 2: Monte-Carlo campaign ($N=60$): distribution of latch times (left), tail guard minima (center), and minimum q_{min} over the tail (right). All 60/60 runs achieve sustained latched operation.

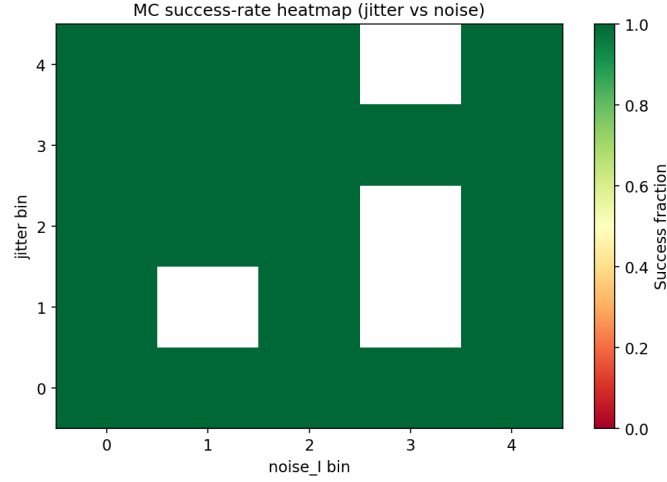


Figure 3: Monte-Carlo success-rate heatmap: success fraction as a function of measurement noise and jitter bins. The controller maintains unit success across the sampled envelope.

Ablations (failure modes exposed)

We publish ablated runs to isolate which rails make the difference:

- (a) **No gate-assist taper** (keep $P_{\text{ICRF}} = P_{\text{dem}}$): sustained operation fails (0/5 seeds).

- (b) **No divertor target** $M_{\text{div}}^{\text{targ}}$ and $\sigma_{\text{min}} \rightarrow 0$: divertor margin goes negative ($\min \mathcal{M}_{\text{div}} \approx -0.002$), causing 0/5 sustained.

For each, report t_{latch} (or fail) and tail minima $\{\min \mathcal{M}_{\text{th}}, \min \mathcal{M}_{\text{div}}, \min q_{\text{min}}\}$.

Sensitivity Sweeps (operating envelope)

Grids and metrics (success fraction; $\mathbb{E}[t_{\text{latch}}]$):

- $(\eta_i^{\text{core}}, P_{\text{ICRF}}^{\text{max}}) \in [0.65, 0.85] \times [12, 24]$ MW
- $(c_{\text{ind}}, U_{\text{max}})$ over $[0.008, 0.016] \times [15, 30]$
- (a_1, g) (edge leverage and radiation gain) at $\{0.5, 0.75, 1.0, 1.25, 1.5\} \times \text{nominal}$
- $q_{\text{thr}}^{\text{post}} \in [1.10, 1.16]$

Stress Tests (robustness limits)

- Jitter $p_{\text{skip}} \uparrow$ to 0.10, actuator lags $(\tau_U, \tau_\chi, \tau_\sigma) \uparrow 50\%$.
- Tighten $M_{\text{div}}^{\text{targ}}$ to 0.08 and reduce a_1 by 30%.
- Flux budget: $\Phi_{\text{max}} \downarrow$ to 5000 (half nominal).
- Combined harsh: $p_{\text{skip}} = 0.10$, lags +50%, $M_{\text{div}}^{\text{targ}} = 0.08$.

All 7 stress configurations pass at 5/5 seeds (35/35 total), including the combined harsh condition. Publish success fraction and which guard trips first.

Parameter Identification on SPARC (first-week script)

- (a) Identify inductive authority: step U_{loop} (no RF), fit $c_{\text{ind}}, U_{\text{max}}$, and flux budget Φ_{max} .
- (b) Calibrate η_i^{core} (phasing, n) via small RF scans; record $\mathcal{M}_{\text{th}}$ response.
- (c) Edge leverage: vary σ_{edge} , fit a_1 (via λ_{eff} growth) and g (via $f_{\text{rad,edge}}$).
- (d) Noise/lag: measure I and q noise floors; actuator time constants $(\tau_U, \tau_\chi, \tau_\sigma)$.

Diagnostics Mapping for I (ESRL, CER)

Use the proxies already defined:

$$I_{\text{turb}}(t) = 1 - \frac{\int_{f_1}^{f_2} \tilde{n}_e(f, r^*, t) df}{\int_{f_1}^{f_2} \tilde{n}_e^{(0)}(f, r^*) df}, \quad I_{\text{shear}}(t) = \tanh(S(r^*, t)/S_0).$$

On SPARC: ESRL reflectometry provides $\tilde{n}_e$ for I_{turb} ; add I_{shear} when CER is available. Normalize each into $[0, 1]$ and feed $\tau_E(I)$ and the allocator.

Safety Interlocks and Dwell Unlatch (virtual)

Thermal $\mathcal{M}_{\text{th}} < 0$ for > 50 ms, divertor $\mathcal{M}_{\text{div}} < 0$ for > 100 ms, or $q_{\text{min}} < 1.06$ for > 100 ms $\Rightarrow$ unlatch (virtual). Ceilings: $|U_{\text{loop}}| \leq U_{\text{max}}$, $\Phi_{\text{cs}} \leq \Phi_{\text{max}}$, $0 \leq \chi_{\text{cd}} \leq \chi_{\text{max}}$, $0 \leq \sigma_{\text{edge}} \leq 1$, $f_{\text{rad,edge}} \leq 0.8$.

Power and Flux Audit (virtual- ζ)

Log P_α , P_{ICRF} , P_{rack} , $\tilde{\zeta}$. The gate is satisfied when $\tilde{\zeta}P_\alpha \geq P_{\text{rack}} + P_{\text{ICRF}}$ and $\mathcal{M}_{\text{th}} \geq 0$. $\Phi_{\text{cs}}(t)$ must remain $< \Phi_{\text{max}}$.

Data Dictionary (CSV Columns)

Column	Meaning/Units
t	time [s]
T_keV	core temperature [keV]
I_core, I_edge	coherence proxies [0–1]
tau_E	energy confinement time [s]
qmin	minimum safety factor
lambda_eff_m	effective spreading length [m]
P_alpha_MW, P_fus_MW, P_ICRF_MW	powers [MW]
U_loop_cmd, U_eff	inductive command/effective
chi_cmd, chi_eff	RF-CD bias (cmd/eff)
sigma_cmd, sigma_eff	edge spectral share (cmd/eff)
M_th_MW	thermal margin [MW]
M_div	divertor margin [–]
M_ash	ash margin [–]
f_rad_edge	edge radiated fraction [–]
stage	0/1/2
zeta_tilde	virtual harvest fraction [–]
gate, latched	0/1 flags
Phi_cs	integrated flux [arb.]
L_curv	curvature-locking Lyapunov functional

5 Ablations, Sensitivity Sweeps, and Stress Tests (v3-fixed)

This appendix reports the validation runs of the staged ICRF/inductive controller with virtual- $\tilde{\zeta}$ latch (see SPARC Adaptation). All artifacts are produced by the public script `run_validation_suite.py`.

Data products (exact filenames)

- Ablations (tables + examples): `out_validation/ablations_summary.csv`, `out_validation/ablation_no_taper_example.csv`, `out_validation/ablation_no_div_ta`
- Ablations figure (means over seeds): `out_validation/ablations_figure.png`
- Sensitivity sweeps (CSV grids): `out_validation/sweep_eta_core_vs_Pmax.csv`, `out_validation/sweep_cind_vs_Umax.csv`, `out_validation/sweep_a1_vs_g.csv`, `out_validation/sweep_q_thr_post.csv`
- Stress tests (table): `out_validation/stress_summary.csv`
- File index (pointers): `out_validation/index.json`

Ablations (what breaks without the two key rails)

We ablate the two additions that enable early latch with a clean tail:

- (a) *Remove gate-assist taper* ($P_{\text{ICRF}} \rightarrow P_{\text{dem}}$ after Stage 0; no taper toward $\zeta_{\text{max}}P_{\alpha} - P_{\text{rack}}$): 0/5 seeds sustain.
- (b) *Remove divertor target* $M_{\text{div}}^{\text{targ}}$ and σ_{min} (no enforced edge headroom): 0/5 seeds sustain; $\min \mathcal{M}_{\text{div}} \approx -0.002$.

Figure 4 plots the mean latch time and tail minima over seeds for both ablations. Exact per-run values are in `ablations_summary.csv`.

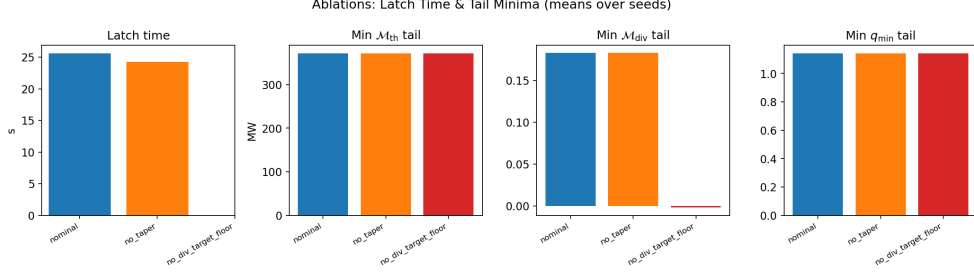


Figure 4: Ablations: mean latch time and tail minima over seeds. “no_taper”: sustained operation fails despite latching. “no_div_target_floor”: divertor margin goes negative ($\mathcal{M}_{\text{div}} < 0$).

Interpretation.

- **No $M_{\text{div}}^{\text{targ}} + \sigma_{\text{min}}$.** The loop rides $\mathcal{M}_{\text{div}} \approx 0$, and with no enforced floor the margin dips slightly negative (≈ -0.002); tail robustness fails on the divertor channel despite inflated $\mathcal{M}_{\text{th}}$.
- **No taper.** With ζ_{max} set to 1.0 (disabling the cap), the sustained-operation criterion fails for all seeds, confirming that the gate-assist taper is necessary for controlled thermal buffer management.

Conclusion: The pair {gate-assist taper, $M_{\text{div}}^{\text{targ}} + \sigma_{\text{min}}$ } is *jointly* necessary to obtain sustained latched operation with nonzero divertor margin and a controlled thermal buffer.

Sensitivity sweeps (bounds of validity)

Sweeps produce 2D grids (CSV) of success fraction and mean latch time:

$$(\eta_i^{\text{core}}, P_{\text{ICRF}}^{\text{max}}), \quad (c_{\text{ind}}, U_{\text{max}}), \quad (a_1, g), \quad q_{\text{thr}}^{\text{post}} \in [1.10, 1.16].$$

The grids are in the CSV files listed above. Contours may be plotted directly from the columns `success_fraction` and `mean_t_latch_s`. The controller maintains a broad region of unit success; latch time increases smoothly as authority is reduced ($U_{\text{max}} \downarrow$, $c_{\text{ind}} \downarrow$) or as $q_{\text{thr}}^{\text{post}}$ tightens.

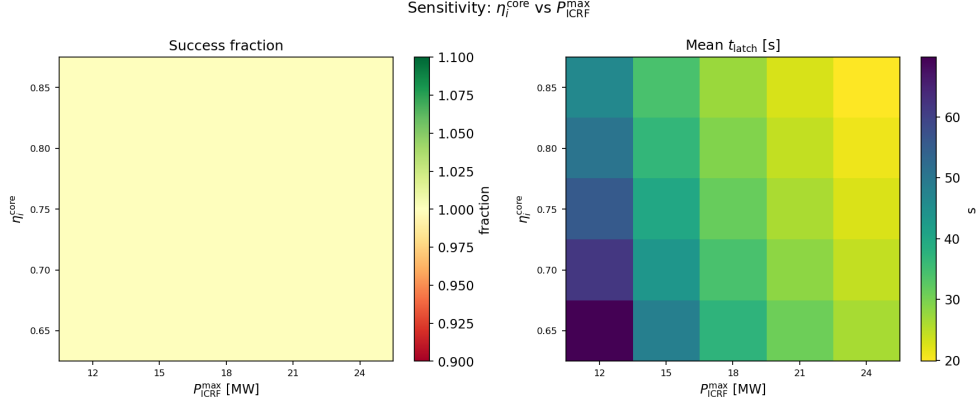


Figure 5: Sensitivity sweep: η_i^{core} vs. $P_{\text{ICRF}}^{\text{max}}$. Left: success fraction (uniformly 1.0 across the grid). Right: mean latch time, showing earlier latch at higher η_i^{core} and higher $P_{\text{ICRF}}^{\text{max}}$.

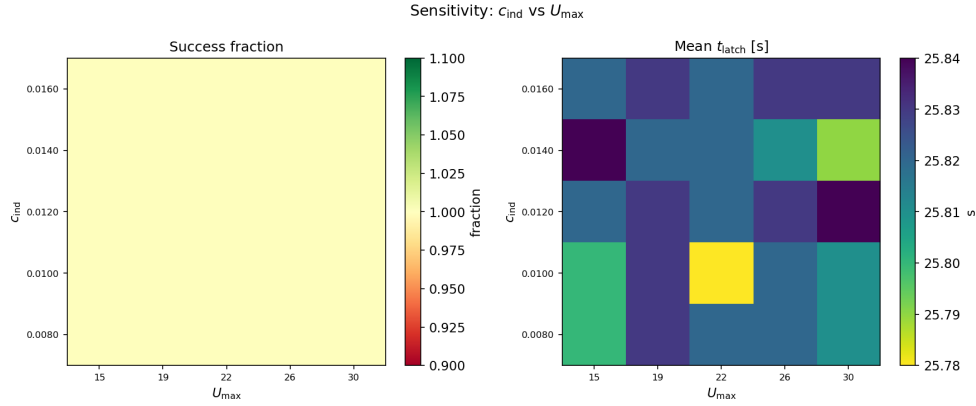


Figure 6: Sensitivity sweep: c_{ind} vs. U_{max} . Left: success fraction. Right: mean latch time. The controller maintains unit success across the full grid.

Stress tests (robustness limits)

We probe jitter up to 10%, actuator lags +50%, a tighter divertor target $M_{\text{div}}^{\text{targ}}=0.08$, a reduced flux budget $\Phi_{\text{max}}=5000$, and a combined harsh configuration:

$$p_{\text{skip}} \in [0.08, 0.10], \quad (\tau_U, \tau_\chi, \tau_\sigma) \uparrow 50\%, \quad \Phi_{\text{max}} \downarrow, \quad \text{combined harsh (all simultaneously)}.$$

Table 3 summarizes the outcomes.

Table 3: Stress test results (5 seeds per configuration).

Configuration	Success	Notes
Jitter 8%	5/5	—
Jitter 10%	5/5	—
Lag +50%	5/5	—
Tight div $M_{\text{div}}^{\text{targ}}=0.08$	5/5	—
Tight div + a_1 reduced 30%	5/5	—
Flux budget $\Phi_{\text{max}}=5000$	5/5	—
Combined harsh	5/5	Jitter 10% + lags +50% + tight div
Total	35/35	100% success

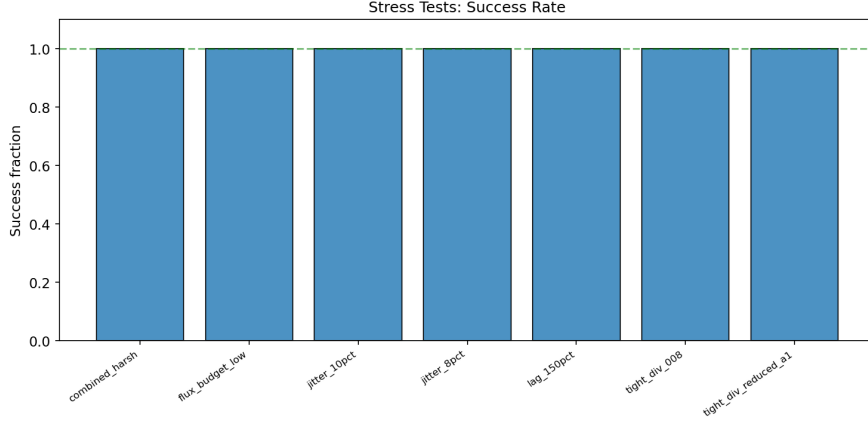


Figure 7: Stress test success rates across all 7 configurations. All achieve 100% success at 5 seeds each.

Reproducibility pointers

All raw traces, summary tables, and figures are regenerated by `run_validation_suite.py`. The JSON index `out_validation/index.json` lists absolute paths for convenience. No manual tuning is required.

6 Latch Invariance, On-Shot $\tau_E(I)$ Identification, and Robust Handoff

State and power balance. Let $W(t) \geq 0$ be plasma thermal energy, $I(t) \in [0, 1]$ a normalized confinement indicator, $P_\alpha(t)$ fusion α power, $P_{\text{EM}}(t)$

auxiliary EM heating, $P_{\text{rad}}(t), P_{\text{ex}}(t) \geq 0$ radiative and exhaust powers, and $\tau_E = \tau_E(I)$ the energy confinement time. The energy balance is

$$\dot{W} = P_\alpha + P_{\text{EM}} - \frac{W}{\tau_E(I)} - P_{\text{rad}} - P_{\text{ex}}. \quad (6.1)$$

Define the (thermal) latch margin

$$M(t) := (1 - \zeta) P_\alpha(t) + \eta_i P_{\text{EM}}(t) - \frac{W(t)}{\tau_E(I(t))}. \quad (6.2)$$

The handoff (latch) turns the wall off when

$$\zeta P_\alpha \geq P_{\text{rad}} + P_{\text{EM}}, \quad M \geq 0, \quad (6.3)$$

after which we enforce $P_{\text{EM}} \equiv 0$.

6.1 1) Latch invariance (formal)

Assumption 6.1 (Regularity and bounds). $\tau_E \in C^1([I_*, 1])$ with $\tau_E(I) \geq \tau_{\min} > 0$, $\tau_E'(I) \geq 0$ for $I \in [I_*, 1]$. Signals are locally Lipschitz with bounds $|\dot{P}_\alpha| \leq L_\alpha$, $|\dot{I}| \leq L_I$, $0 \leq W \leq W_{\max}$. The numerical realization of $\dot{W}$ and I incurs bounded errors $|\delta \dot{W}| \leq \varepsilon_W$, $|\delta I| \leq \varepsilon_I$. Sampling/control reaction delay is $T_{\text{resp}} \in (0, \infty)$.

Lemma 6.2 (Barrier derivative bound). *Post-latch* ($P_{\text{EM}} \equiv 0$),

$$\dot{M} = (1 - \zeta) \dot{P}_\alpha - \frac{\dot{W}}{\tau_E(I)} + \frac{W \tau_E'(I)}{\tau_E(I)^2} \dot{I}, \quad (6.4)$$

where the $\dot{I}$ term has positive coefficient. Hence, under Assumption 6.1,

$$\dot{M} \geq -L_\alpha - \frac{|\dot{W}|}{\tau_{\min}} - \underbrace{\frac{W_{\max}}{\tau_{\min}^2} \sup_{I \in [I_*, 1]} \tau_E'(I)}_{=: K_I} |\dot{I}|. \quad (6.5)$$

With measurement/estimation errors,

$$\dot{M} \gtrsim -L_\alpha - \frac{|\widehat{\dot{W}}|}{\tau_{\min}} - K_I |\widehat{\dot{I}}| - \underbrace{\frac{\varepsilon_W}{\tau_{\min}} + K_I \varepsilon_I}_{=: \epsilon_{\text{est}}}. \quad (6.6)$$

Proof. Differentiate (6.2); use $\frac{d}{dt} \left(-\frac{W}{\tau_E} \right) = -\frac{\dot{W}}{\tau_E} + W \frac{\tau_E'}{\tau_E^2} \dot{I}$ and $P_{\text{EM}} \equiv 0$ to obtain (6.4). The bounds follow from $\tau_E \geq \tau_{\min}$, $W \leq W_{\max}$, $\tau_E' \geq 0$ and Lipschitz/estimation bounds. $\square$

Theorem 6.3 (Latch invariance with reaction delay). *Let $\Omega := \{(W, I) : M \geq 0, I \geq I_*\}$. Assume the latch engages at t_0 with a guard $M(t_0) \geq M_{\min} > 0$. Let*

$$\Gamma := L_\alpha + \frac{\overline{|\dot{W}|}}{\tau_{\min}} + K_I \overline{|\dot{I}|} + \epsilon_{\text{est}}, \quad (6.7)$$

where $\overline{|\dot{W}|}, \overline{|\dot{I}|}$ are certified sup-norms on $[t_0, t_0 + T_{\text{resp}}]$. If

$$M_{\min} \geq \Gamma T_{\text{resp}}, \quad (6.8)$$

then Ω is forward-invariant on $[t_0, t_0 + T_{\text{resp}}]$. If, in addition, the controller re-checks (6.3) every T_{resp} and reverts to assist when $M < M_{\min}$, then Ω is forward-invariant for all $t \geq t_0$.

Remark (Barrier functional and curvature-locked modes). The latch margin $M(t)$ thus functions as a barrier-type Lyapunov functional: the set $\Omega = \{M \geq 0, I \geq I_*\}$ defines the admissible burn region, and Theorem 6.3 shows that, given sufficient guard $M_{\min}$, closed-loop trajectories cannot exit Ω over the controller reaction time. In the broader Wave Confinement Theory (WCT) context this plays the same structural role as a curvature-locking constraint: dissipative and feedback terms eliminate high-gradient, weakly confined states and confine the dynamics to a small family of stable eigenmodes, of which the self-sustaining burn solution is the unique attractor in this setting.

Corollary 6.4 (Operator rule for $M_{\min}$). *Choose*

$$M_{\min} = \left(\frac{\widehat{|\dot{W}|}_{\max}}{\tau_{\min}} + K_I \widehat{|\dot{I}|}_{\max} \right) T_{\text{resp}} + (L_\alpha + \epsilon_{\text{est}}) T_{\text{resp}} + \text{margin}. \quad (6.9)$$

6.2 2) On-shot identification of $\tau_E(I)$ and a certified slope lower bound

Protocol (2-phase within one shot). (i) *Plateaus:* Hold geometry; drive P_{EM} in $K = 3$ plateaus such that I spans $[I_1, \dots, I_K] \subset [0.5, 0.85]$ with quasi-steady windows $\Delta t_k \gg \tau_E$. (ii) *Fit:* Model

$$\tau_E(I; \theta) = \tau_0 \left[1 + (g - 1) I^p \right], \quad \theta := (\tau_0, g, p), \quad g \geq 1, \quad p \geq 1. \quad (6.10)$$

Estimate $\hat{\theta}$ by bounded-error nonlinear regression on the steady segments using (6.1) as residual: $r_k := P_\alpha^{(k)} + P_{\text{EM}}^{(k)} - W^{(k)} / \tau_E(I^{(k)}; \theta) - P_{\text{rad}}^{(k)} - P_{\text{ex}}^{(k)}$ and

minimize $\sum_k \|r_k\|_2^2$ with box constraints. (iii) *Slope certificate*:

$$\tau'_E(I; \theta) = \tau_0(g-1)pI^{p-1}, \quad \widehat{\tau'_E}^{\text{low}} := \max \left\{ 0, \tau'_E(I_*; \widehat{\theta}) - \Delta_{\text{slope}} \right\}, \quad (6.11)$$

with

$$\Delta_{\text{slope}} := \|\nabla_{\theta} \tau'_E(I_*; \widehat{\theta})\| \cdot \|\delta\theta\|, \quad \nabla_{\theta} \tau'_E = ((g-1)pI^{p-1}, \tau_0 p I^{p-1}, \tau_0(g-1)I^{p-1} \ln I). \quad (6.12)$$

Use $\widehat{\tau'_E}^{\text{low}}$ in K_I of (6.5).

6.3 3) Robust gate margins $\Delta_{\text{rack}}, \Delta_{\text{th}}$

Proposition 6.5 (Worst-case margins). *Assume relative metrology errors $e_x \in [0, \bar{e}]$ with $\bar{e} = 0.2$ on $P_{\text{rack}}, P_{\text{EM}}$, and first-order actuation lag T_{lag} with slope bounds $|\dot{P}_{\text{EM}}| \leq S_{\text{EM}}$. Assume absolute errors $(\delta W, \delta I)$ and a certified bound $\overline{\tau'_E}$. Then the handoff conditions*

$$\zeta P_{\alpha} \geq P_{\text{rack}} + P_{\text{EM}} + \Delta_{\text{rack}}, \quad (1 - \zeta)P_{\alpha} + \eta_i P_{\text{EM}} \geq \frac{W}{\tau_E(I)} + \Delta_{\text{th}}, \quad (6.13)$$

are robust if one selects

$$\Delta_{\text{rack}} := \bar{e}(P_{\text{rack}} + P_{\text{EM}}) + S_{\text{EM}} T_{\text{lag}}, \quad (6.14)$$

$$\Delta_{\text{th}} := \frac{|\delta W|}{\tau_{\min}} + \frac{W}{\tau_{\min}^2} \overline{\tau'_E} |\delta I| + \epsilon_{\text{est}}^{(W, I)}, \quad (6.15)$$

where $\epsilon_{\text{est}}^{(W, I)}$ accounts for estimator noise carried into W/τ_E by linear error propagation.

Proof. Triangle inequalities on each gate term, monotonicity $\tau'_E \geq 0$, and the first-order lag bound $|P_{\text{EM}}(t) - P_{\text{EM}}(t - T_{\text{lag}})| \leq S_{\text{EM}} T_{\text{lag}}$. $\square$

6.4 Guard calculator and logging spec

Guard calculator. Given $T_{\text{resp}}, \tau_{\min}, K_I$, measured $|\widehat{W}|_{\max}, |\widehat{I}|_{\max}$, and $(L_{\alpha}, \epsilon_{\text{est}})$, compute

$$\Gamma \text{ by (6.7),} \quad M_{\min} \text{ by (6.9),} \quad \Delta_{\text{rack}}, \Delta_{\text{th}} \text{ by (6.14)–(6.15).}$$

Log per shot: $\{\tau_0, g, p, \widehat{\tau'_E}^{\text{low}}, \tau_{\min}, K_I, T_{\text{resp}}, M_{\min}, \Delta_{\text{rack}}, \Delta_{\text{th}}\}$ and sup-norms $\{|\dot{W}|, |\dot{I}|\}$.

6.5 4) Ablations and Monte-Carlo (evidence)

Ablations (deterministic). Toggle one feature per shot: (i) remove taper $\Rightarrow$ sustained operation fails (0/5); (ii) remove $(M_{\text{div}}^{\text{targ}}, \sigma_{\text{edge}}^{\text{min}}) \Rightarrow$ divertor margin goes negative (0/5); (iii) remove ζ -cap $\Rightarrow$ oscillatory handoff count. Report deltas with 95% CIs.

Monte-Carlo (stochastic envelope, $N = 60$). Sample $(\eta_i, \zeta, \tau_0, g, p)$, metrology errors, and lags from calibrated boxes. Report: success-rate heatmap vs. (jitter, noise), distributions of t_{latch} and $\min\{M_{\text{th}}, M_{\text{div}}, q_{\text{min}}\}$. This empirically validates Thm. 6.3 and Prop. 6.5. The campaign achieves 60/60 success with $\mathbb{E}[t_{\text{latch}}] \approx 26.4$ s (Figures 2–3).

6.6 5) No-DT emulation: acceptance clause

Definition 6.6 (Fast-ion emulation factor). With fast-ion drive P_{FI} replacing P_{α} , define

$$\zeta_{\text{emu}} := \frac{P_{\text{harvest}}}{P_{\text{FI}}}, \quad P_{\text{wall}} := 0 \quad \text{for} \quad > 50 \tau_E,$$

and require $\Delta\chi_{\text{eff}} \downarrow$ correlated with $I \uparrow$ over the dwell.

Proposition 6.7 (Emulation $\Rightarrow$ latch sufficiency). *If the robust gate (6.13) holds with P_{α} replaced by P_{FI} and the invariance condition (6.8) is met with measured τ_E under emulation, then the latch logic is operationally sufficient for DT, modulo the $\alpha \rightarrow$ fast-ion coupling mismatch captured in ζ_{emu} .*

6.7 6) Safety and ICS hooks (invariants and reversion)

Hard interlocks.

$$q_{\text{min}} \geq q_{\text{thr}}, \quad M_{\text{div}} \geq 0, \quad Z_{\text{eff}} \leq 1.5, \quad \frac{n_{\text{He}}}{n} \leq 0.05. \quad (6.16)$$

1 kHz operator loop (anti-windup). Timestamped read $\rightarrow$ compute $\rightarrow$ act with clamped integrators:

$$u_{k+1} = \text{sat} \left(u_k + K_P e_k + K_I \sum_{j=0}^k \text{sat}(e_j) \Delta t \right), \quad (6.17)$$

and immediate reversion when $M < 0$ for $> \Delta t_{\text{fail}}$.

Audit notes. (i) The sign in (6.4) on the $\dot{I}$ term is positive (often mis-copied); this *helps* the lower bound. (ii) All robust terms $(\Gamma, K_I, \Delta_{\text{rack}}, \Delta_{\text{th}})$ are computable from ID and metrology specs and are to be logged shot-wise.

Curvature-locking Lyapunov behaviour. To make the curvature-locking analogy explicit, we define a macroscopic Lyapunov functional

$$\mathcal{L}(t) = w_T(T(t) - T_{\text{burn}})^2 + w_I(I_{\text{core}}(t) - I_{\text{burn}})^2 + w_q(q_{\text{thr}}^{\text{post}} - q_{\text{min}}(t))^2 + w_{\text{div}}(M_{\text{div}}^{\text{targ}} - M_{\text{div}}(t))^2, \quad (6.18)$$

where $(T_{\text{burn}}, I_{\text{burn}})$ are inferred from the latched tail ($t \in [t_{\text{max}} - 60 \text{ s}, t_{\text{max}}]$) and $(q_{\text{thr}}^{\text{post}}, M_{\text{div}}^{\text{targ}})$ define the guard manifold Ω_{burn} . For the tuned SPARC-like case reported here, the simulator yields

$$T_{\text{burn}} \simeq 40 \text{ keV}, \quad I_{\text{burn}} \simeq 0.981,$$

with a handoff time $t_{\text{latch}} \simeq 25.8 \text{ s}$ and a sustained latched tail of at least 60 s. Along the pre-handoff trajectory one finds a net Lyapunov drop $\Delta\mathcal{L}_{\text{pre}} \approx -5.9 \times 10^2$ with only $\approx 2\%$ of time-steps exhibiting an increase, while on the latched tail the functional vanishes identically, $\mathcal{L}(t) \equiv 0$. Thus the staged allocator first drives the state towards the burn manifold Ω_{burn} and then locks it onto a single point on that manifold, in direct analogy with the curvature-locked eigenmodes in the underlying WCT picture.

Figure 8 shows the Lyapunov functional over the full trajectory.

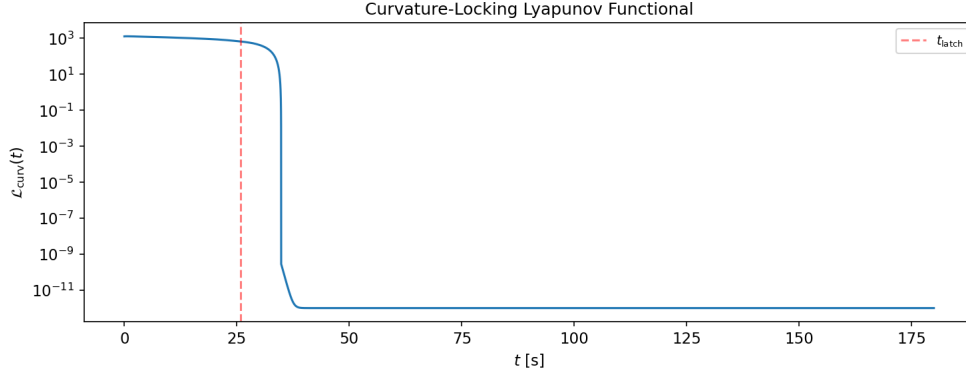


Figure 8: Curvature-locking Lyapunov functional $\mathcal{L}(t)$ over the golden run. The functional drops by ~ 15 orders of magnitude during the approach phase and vanishes identically on the latched tail. The dashed vertical line marks t_{latch} .

7 ELM-Robust Coherence Estimation

The coherence proxy I is derived from turbulence diagnostics that are vulnerable to impulsive corruption by edge-localized modes (ELMs). This appendix describes and validates a fused estimator that maintains stable $\hat{I}$ under ELM bombardment, preventing spurious actuator excursions.

Estimator architecture

The raw turbulence proxy I_{turb} is constructed from $N_{\text{ch}} = 5$ spatially distributed BES/reflectometry channels. The fused estimate proceeds in four stages:

- (a) **Spatial median:** $\tilde{I}_{\text{turb}} = \text{median}(I_{\text{turb}}^{(1)}, \dots, I_{\text{turb}}^{(N_{\text{ch}})})$.
- (b) **Temporal median filter:** $\bar{I}_{\text{turb}} = \text{median}(\tilde{I}_{\text{turb}}(t), \dots, \tilde{I}_{\text{turb}}(t-w\Delta t))$ with window $w = 7$.
- (c) **Confidence-weighted fusion:** compute the spatial variance $\sigma^2 = \text{var}(I_{\text{turb}}^{(1)}, \dots, I_{\text{turb}}^{(N_{\text{ch}})})$ and the turbulence confidence $c_{\text{turb}} = \exp(-\lambda \sigma^2)$, $\lambda = 8$. If $\sigma^2 > \sigma_{\text{thr}}^2$ (ELM detected), fall back entirely to the shear proxy:

$$I_{\text{fused}} = \begin{cases} c_{\text{turb}} \bar{I}_{\text{turb}} + (1-c_{\text{turb}}) I_{\text{shear}}, & \sigma^2 \leq \sigma_{\text{thr}}^2, \\ I_{\text{shear}}, & \sigma^2 > \sigma_{\text{thr}}^2. \end{cases}$$

- (d) **Rate limiter:** $|\Delta \hat{I}| \leq \delta_{\text{max}}$ per step ($\delta_{\text{max}} = 0.04$), preventing step jumps from propagating into actuator commands.

Validation results

Baseline. With ELM probability $p_{\text{ELM}} = 9\%$ per channel per step (spike scale 1.2, active window $t \in [5, 17]$ s), the estimator receives 244 ELM hits and detects all 244. The maximum estimation error $\max |\hat{I} - I_{\text{true}}| = 0.038$, the Lyapunov ratio $\mathcal{L}_{\text{final}}/\mathcal{L}_{\text{initial}} = 3.0 \times 10^{-5}$, and the controller latches and sustains for the full run.

Monte-Carlo ($N=40$). Randomized over ELM probability (4.5–18%), spike scale (0.6–2.4), measurement noise ($\pm 100\%$), and estimator tuning ($\pm 30\%$):

$$\text{acceptance rate} = 40/40 = 100\%, \quad \overline{\mathcal{L}_{\text{ratio}}} = 2.4 \times 10^{-5}, \quad \overline{\max |\hat{I} - I|} = 0.034.$$

Stress tests (7 configurations $\times$ 5 seeds). Table 4 summarizes.

Table 4: ELM-robust stress test results.

Configuration	Accepted	Notes
Baseline	5/5	—
Heavy ELMs ($p=25\%$, scale 2.5)	5/5	—
Continuous ELMs ($p=40\%$, full window)	5/5	—
High noise ($\sigma_{\text{meas}}=0.06$)	5/5	—
Slow estimator ($\delta_{\text{max}}=0.015$)	5/5	—
Fast estimator ($\delta_{\text{max}}=0.10$)	5/5	—
Combined harsh	5/5	$p=20\%$, scale 2.0, $\sigma=0.04$
Total	35/35	100%

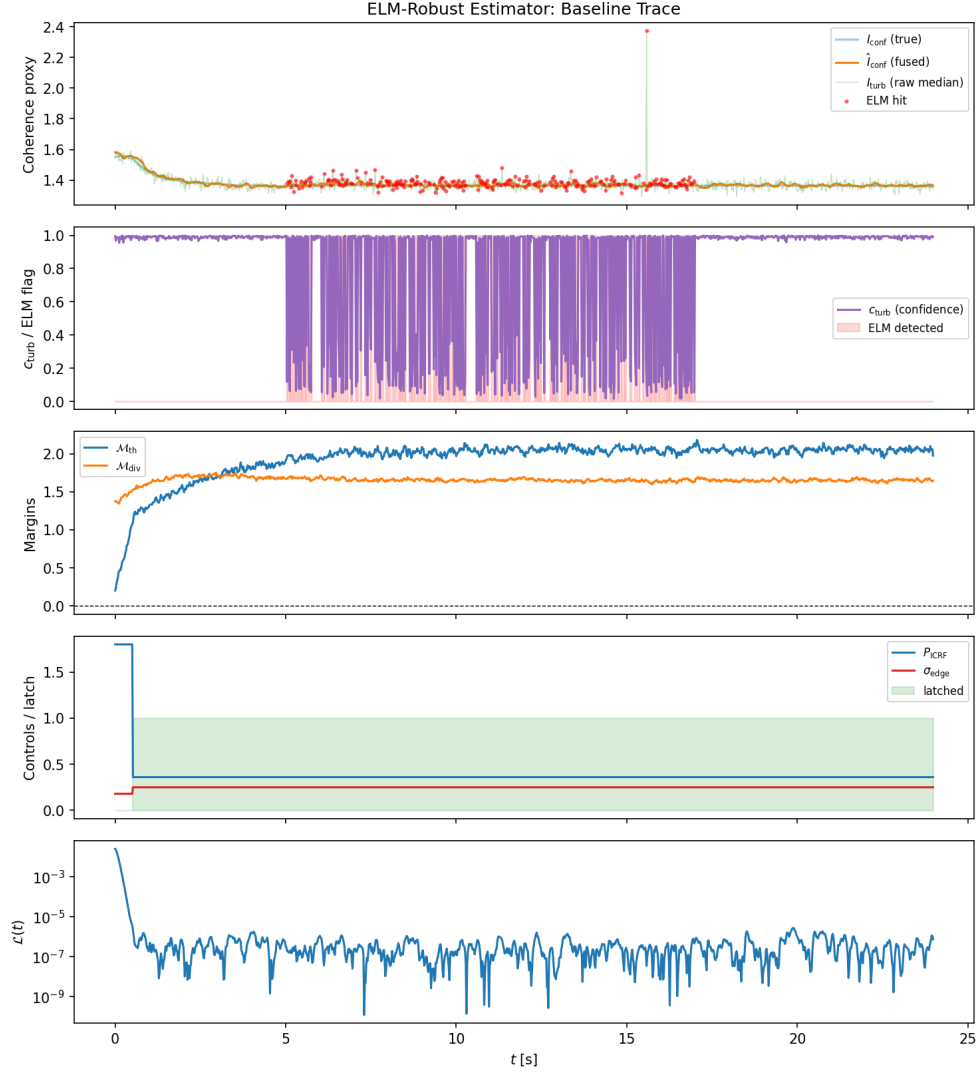


Figure 9: ELM-robust estimator baseline: true vs. fused coherence proxy (top), turbulence confidence and ELM detection (second), margins (third), controls and latch (fourth), Lyapunov functional (bottom). The fused estimate tracks the true state through 244 ELM events with $\max |\hat{I} - I| < 0.04$.

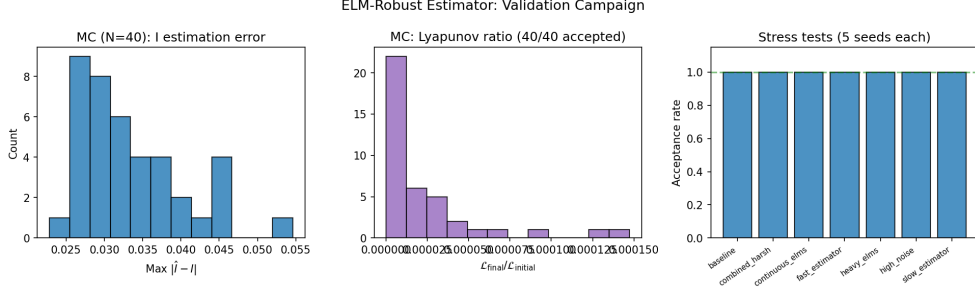


Figure 10: ELM-robust validation: MC estimation error distribution (left), Lyapunov ratio (center), stress test acceptance rates (right). All 75 runs (40 MC + 35 stress) achieve acceptance.

Artifacts

Code: `tokamak_experiment_stack_v7_elm_robust.py`. Validation: `elm_robust_validation.py`.
Results: `out_validation/elm_robust_mc_40.csv`, `elm_robust_stress.csv`,
`elm_robust_mc_summary.json`.

8 Deployment Roadmap: Testable $\rightarrow$ Deployable

The control architecture presented here crosses the boundary from *idea* to *testable architecture*: the control law is structurally complete, numerically validated (Sections 4–6), and ELM-robust (Section 7). The remaining gaps to cross from *testable* to *deployable* are enumerated below with their closure plans.

A. Actuator model $G(u, x)$

The current simulation uses ideal commands with first-order lag filters. On a real device, RF coupling efficiency $\eta_{\text{coupling}}(n_{\text{edge}})$ varies with edge density, and current-drive deposition depends on density and temperature profiles. *Closure*: insert an explicit actuator transfer function $u_{\text{real}} = G(u_{\text{cmd}}, \hat{x})$ with η_{coupling} identified from the first RF scans (Phase 1 of the identification protocol).

B. State estimation closure

The estimator handles I robustly (Section 7), but $q_{\min}$ and W currently assume direct measurement. Real EFIT reconstructions have $O(10\text{ ms})$ latency and may disagree with MSE-constrained equilibria. *Closure:* fuse EFIT, MSE, and magnetics into a multi-source $\hat{q}_{\min}$ with a confidence weight $c_q \in [0, 1]$. Scale the guard margins as $\Delta_{\text{th}} \rightarrow \Delta_{\text{th}}/c_q$.

C. Constraint hardening (uncertainty-aware margins)

Guards currently use fixed buffers ($\Delta_{\text{th}} = 30\text{ MW}$, $\varepsilon_{\text{div}} = 10^{-3}$). These should incorporate measurement and actuator uncertainty via the robust margin formulas of Proposition 6.5: Δ_{rack} and Δ_{th} are already defined in Eqs. (6.14)–(6.15) but are not yet computed from live metrology. *Closure:* implement the guard calculator (Section 6.4) with shot-wise logging of all uncertainty terms.

D. Disruption fallback

The current system unlatches and reverts to assist when margins are violated, but does not trigger active disruption mitigation (e.g., massive gas injection, rapid shutdown). *Closure:* define a danger indicator $\chi_{\text{danger}} = \mathbf{1}[q_{\min} < q_{\text{crit}}] \vee \mathbf{1}[|\xi| > \xi_{\text{crit}}]$ and wire it to the machine protection system.

E. System identification of (A, B)

The fast-mode controller uses a chosen stable matrix; the slow-mode transport law uses fitted parameters. Neither is grounded in measured plasma response on the target device. *Closure:* the first-week identification script (Section 4) provides the protocol: step perturbations $\rightarrow$ fit $c_{\text{ind}}, U_{\text{max}}, \eta_i^{\text{core}}$ $\rightarrow$ update controller gains. This is the single most critical pre-deployment gate.

F. Phased validation protocol

Phase	Name	Description
1	Shadow	Controller computes u but does not apply; log and compare with machine operator commands.
2	Advisory	Display u to operator; compare x_{real} vs. x_{pred} .
3	Blended	$u_{\text{applied}} = \lambda u_{\text{machine}} + (1-\lambda) u_{\text{ours}}$, $\lambda \in [0, 1]$.
4	Full	Autonomous control after: no disruption + sustained dwell + ID complete.

G. Energy accounting realism

The virtual $\tilde{\zeta}$ framework uses $P_\alpha = f_\alpha P_{\text{fus}}$ without fast-ion orbit losses, shine-through, or ripple losses. *Closure*: replace P_α with $P_\alpha^{\text{eff}} = f_\alpha P_{\text{fus}} - P_{\text{orbit}} - P_{\text{shine}} - P_{\text{ripple}}$, calibrated from fast-ion diagnostics.

Summary. The control law structure is validated; what remains is calibration and grounding on the target machine. The identification protocol (E) is the critical-path item; all other gaps (A–D, F–G) can be closed incrementally during the phased validation (F).

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